



OBSERVE · DIAGNOSE · CONTAIN · FAIL OPEN

Network observability and emergency control, without CNI lock-in.

A product perspective for enterprises that need CNI-independent visibility, stack-deep diagnostics, and leased emergency containment — observe-first, Apache-2.0.

STANDALONE eBPF

KUBERNETES

LINUX

LEASED ENFORCE

OPTIONAL CILIUM

THE NETWORK PROBLEM ENTERPRISES FACE

Clusters grow more complex.

Network certainty does not.

01 Blind spots at the node

CNI dashboards miss host softnet, qdisc, and kernel drop reasons that kill packets before policy.

02 CNI lock-in for visibility

Switching or mixing networks should not mean losing observability and emergency control.

03 Drops without a story

Counters fire; operators still cannot answer where in the stack loss is happening.

04 Dangerous permanent denies

Incident kill-switches that never auto-revert become tomorrow's outage.

05 Tool sprawl

Capture, alerts, SIEM, ChatOps, and AI each live in a different console.

06 Payload risk

Buyers need evidence without collecting application bodies, argv, or secrets.

WHAT NETRA IS

Standalone eBPF observability and leased emergency control.

Netra's node agent owns programs under `/sys/fs/bpf/netra`. Cilium and Hubble are optional enrichment — never a hard dependency. Custom enforcement is time-leased and fails open to observe.

OBSERVE

Flows, DNS/SNI/HTTP host metadata, topology, TCP health — metadata-only, no application payloads.

DIAGNOSE

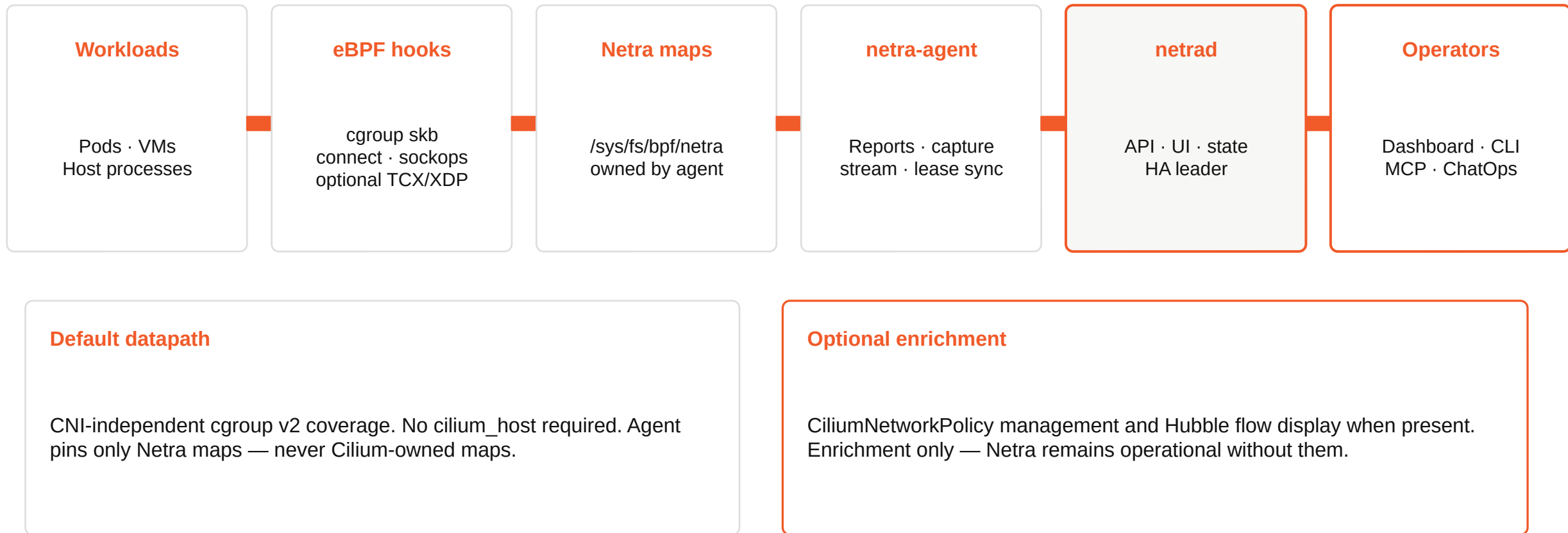
Path, Drop, Congestion Map, ICMP/DNS explain, and filtered packet capture with opt-in auto-PCAP.

CONTAIN

Emergency deny (IP, CIDR, port, DNS, SNI, process, rate) behind an observe/enforce lease that auto-reverts.

HOW NETRA WORKS

From workload to operator — one observe path.



OBSERVE

See traffic as the kernel sees it — metadata only.

L2/L3

IPv4 · IPv6
Direction · hooks

L4

TCP · UDP · ICMP
Flags · ports

L7 meta

DNS name
TLS SNI · HTTP Host

Identity

Namespace · Pod
comm · UID · cgroup

Hard boundary

Netra does not collect application payloads, argv/cmdline, or Secret contents. DNS is ordinary UDP/53 query names. TLS is best-effort ClientHello SNI in a single skb. HTTP is method + Host only — never bodies or paths.

DIAGNOSE

Five surfaces. One investigation story.

01

Path

Connect latency, cwnd pressure, retransmits, edge TCP intel — workload-scoped.

02

Drop

kfree_skb reasons, softnet, interface and qdisc counters — where packets die.

03

Congestion Map

Pictorial stack stages colored by severity across the cluster.

04

Explain

DNS rcode, ICMP MTU/unreachable, next-checks for Health findings.

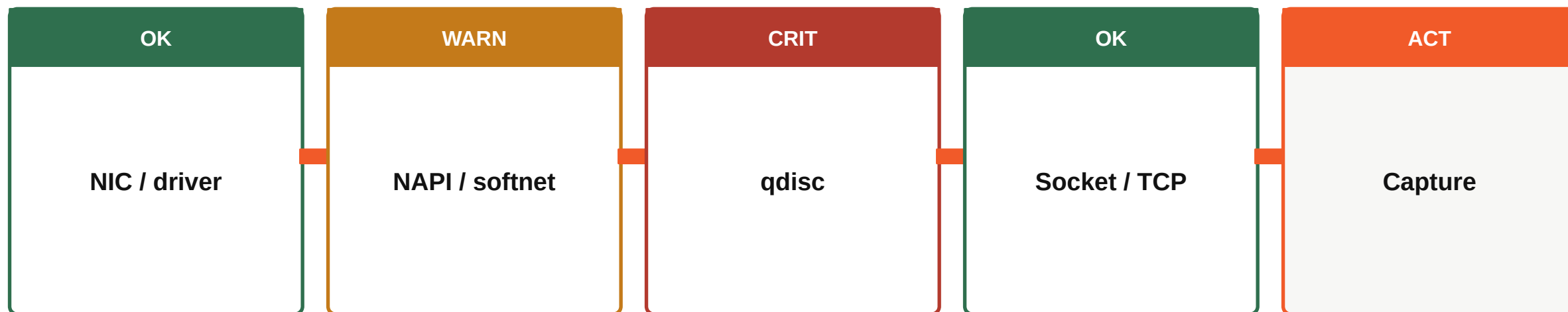
05

Capture

Filtered live capture + opt-in auto-PCAP on critical signals.

CONGESTION MAP — HOW IT WORKS

Follow loss through the Linux stack.



Windowed sysctl and /proc counters color each stage. Critical findings offer one-click Capture — or opt-in auto-capture writes a PCAP automatically.

Netra never writes a sysctl. Guidance is review-only and reversible.

PACKET CAPTURE

Evidence when you need it — filtered and time-bounded.

MANUAL**Operator-started capture**

- 01 Choose node(s), protocol / host / port
- 02 Backend: eBPF (default) or AF_PACKET
- 03 Live stream to dashboard · Wireshark decode
- 04 Export .pcap / .json / .csv · history

AUTO-CAPTURE**Critical signal → PCAP**

- 01 Opt-in NETRA_AUTO_CAPTURE
- 02 Triggers: softnet≥1000, congestion critical, drop spikes
- 03 Agent streams frames → classic PCAP on controller
- 04 History Download · notify event · cooldown

CONTAIN — LEASED EMERGENCY CONTROL

Stop the bleed. Automatically return to observe.

Exact IP / CIDR deny
(+ SYN-drop mode)

Port · DNS · SNI
UID · process comm

PPS / BPS ceilings
Conn-rate per workload

Optional XDP early
ingress on selected NICs

Enforce lease lifecycle

Observe

Default

Plan + confirm

Risk gate

Enforce (leased)

Time-bounded

Auto-revert

Fail open

SAFETY MODEL

Built for production risk tolerance.

01 Observe-first

Enforcement is never the silent default.
Every custom rule rides a lease.

02 Plan + risk confirm

Policy apply stays behind plan-token and
explicit confirmation.

03 Fail open

Lease expiry, agent loss, controller restart,
HA handoff → observe.

04 Blast-radius preview

Review-only deny preview against live
counters before you apply.

05 Map isolation

Netra never pins over Cilium-owned BPF
maps.

06 No secret harvesting

No payloads, argv, or Secret contents in
reports or AI context.

OPERATE & INTEGRATE

One control plane for humans and machines.

Dashboard

HTTPS console — Overview, Firewall, Health, Congestion Map, Capture, Insights.

Alerting

Webhook, email, Slack, Teams, Twilio, HTTP bridge — HA leader-only.

SIEM / metrics

Pull export (JSON/CEF/syslog/OTLP), Prometheus, Grafana, optional Snowflake.

ChatOps

Slack & Teams: status, health, ask — mode always needs confirm.

AI briefs

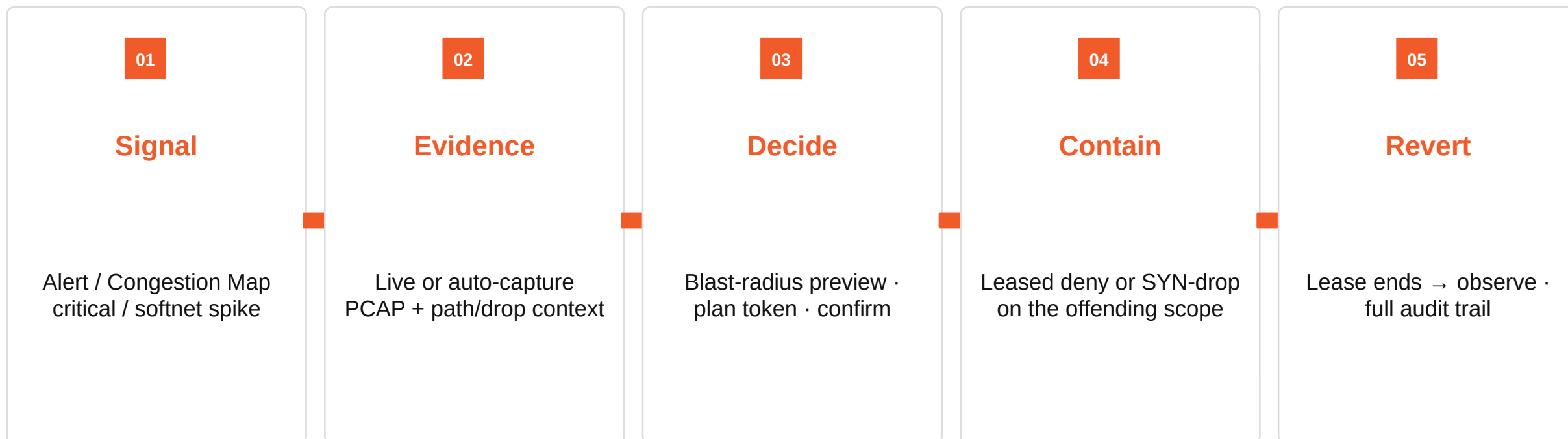
Ask / digest / explain / draft — read-only; never flips enforce.

MCP

133 tools for AI agents; mutations gated off by default.

HOW A DAY LOOKS

From anomaly to contained incident — then safe again.



WHY BUY NETRA

Capability that shows up as control.

CNI-independent

Keep observability when the network stack changes or is mixed.

Fail-open by design

Emergency controls that cannot silently become permanent policy.

Stack-deep diagnostics

Softnet to socket — not just CNI verdicts.

Evidence on demand

Filtered capture and auto-PCAP for critical incidents.

Operator-ready surface

UI, CLI, alerts, SIEM, ChatOps, MCP — one product.

Apache-2.0

Open source core with enterprise operational discipline.

NEXT

See the network.
Contain the incident.
Fail open by default.

[TALK TO SALES →](#)[zyvor.dev/netra →](https://zyvor.dev/netra)

Standalone eBPF network observability and emergency network control for Linux/Kubernetes.